

Skip Module Hyperparameter Selection

1 Algorithm

Algorithm 1 Skip Module $\mathcal{S}$ with Eager Mode and Periodic Forced Check

Require: Same inputs as Algorithm ??, plus forced check interval N_{chk} , clear confirmation window W_{clr}

Ensure: Skip decisions $\{s_1, s_2, \dots\}$, where $s_t \in \{0, 1\}$

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1:  $f_{\text{prev}} \leftarrow \text{null}$ 
2:  $\text{eager\_mode} \leftarrow \text{TRUE}$ 
3:  $\text{non\_firesmoke\_streak} \leftarrow 0$ 
4:  $\text{skip\_streak} \leftarrow 0$ 
5: for each incoming frame  $f_t$  do
    Step 1: Pre-process
6:   (Same as Algorithm ??, Step 1)
    Step 2: Eager Mode Override
7:   if  $\text{eager\_mode} = \text{TRUE}$  then
8:      $s_t \leftarrow 1$ 
9:      $\hat{y}_t \leftarrow \text{RUNDL}(f_t)$ 
10:    if  $\hat{y}_t = \text{fire\_smoke}$  then
11:       $\text{non\_firesmoke\_streak} \leftarrow 0$ 
12:    else
13:       $\text{non\_firesmoke\_streak} \leftarrow \text{non\_fire\_streak} + 1$ 
14:      if  $\text{non\_firesmoke\_streak} \geq W_{\text{clr}}$  then
15:         $\text{eager\_mode} \leftarrow \text{FALSE}$ 
16:         $\text{non\_firesmoke\_streak} \leftarrow 0$ 
17:         $\text{skip\_streak} \leftarrow 0$ 
18:      end if
19:    end if
20:     $f_{\text{prev}} \leftarrow \tilde{f}_t$ 
21:    continue
22:  end if
    Step 3: Periodic Forced Check
23:    if  $\text{skip\_streak} \geq N_{\text{chk}}$  then
24:       $s_t \leftarrow 1$ 
25:       $\hat{y}_t \leftarrow \text{RUNDL}(f_t)$ 
26:       $\text{skip\_streak} \leftarrow 0$ 
27:      if  $\hat{y}_t = \text{fire\_smoke}$  then
28:         $\text{eager\_mode} \leftarrow \text{TRUE}$ 
29:         $\text{non\_firesmoke\_streak} \leftarrow 0$ 
30:      end if
31:       $f_{\text{prev}} \leftarrow \tilde{f}_t$ 
32:      continue
33:    end if
    Step 4: Compute Foreground Mask
34:    (Same as Algorithm ??, Step 2)
    Step 5: Gate Decision
35:    if any active block exists then
36:       $s_t \leftarrow 1$ ;  $\hat{y}_t \leftarrow \text{RUNDL}(f_t)$ 
37:       $\text{skip\_streak} \leftarrow 0$ 
38:      if  $\hat{y}_t = \text{fire\_smoke}$  then
39:         $\text{eager\_mode} \leftarrow \text{TRUE}$ 
40:         $\text{non\_firesmoke\_streak} \leftarrow 0$ 
41:      end if
42:    else
43:       $s_t \leftarrow 0$ 
44:       $\text{skip\_streak} \leftarrow \text{skip\_streak} + 1$ 
45:    end if
46:     $f_{\text{prev}} \leftarrow \tilde{f}_t$ 
47:    yield  $s_t$ 
48: end for
```

▷ Start in eager mode — skipping must be earned
▷ Must reach W_{clr} before first skip is allowed
▷ Consecutive skipped frames — forced check trigger
(new)
▷ Skip disabled — always run DL inference
▷ Fire still present — stay in eager mode
▷ Scene confirmed clear — exit eager mode
(new)
▷ Force-run DL — motion gate overridden
▷ Forced check detected fire — enter eager mode
▷ Motion gate detected fire — enter eager mode
▷ No motion — skip
